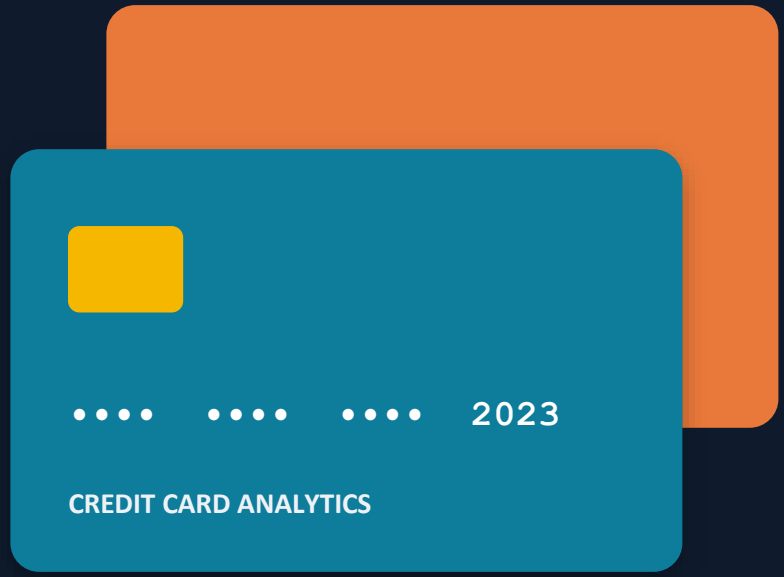
Credit Card Financial Dashboard

Revenue, transaction and customer analytics by year, quarter and month, built with Power BI and DAX

Power BI | DAX | PostgreSQL



Project objective

Give stakeholders an interactive view of credit card revenue, interest, transactions and customer segments, filterable by year, quarter, month and week, so trends can be spotted and acted on quickly.

Questions the dashboard answers

- Which cards, channels and spend types drive revenue?
- Which customer segments (age, income, state) are most valuable?
- How is revenue trending across quarters, months and years?

55.3M
Total revenue

7.8M
Interest earned

656K
Transactions

10,108
Customers

From raw data to dashboard

CSV to PostgreSQL
Tables created in the ccdb database, then CSV data loaded with COPY (10,108 rows each)

Power BI import
Connected to PostgreSQL in Import mode; Date_Table generated with DAX

Data model
One-to-one on client_num between customers and cards; calendar joined one-to-many

DAX layer
Calculated columns for segments and revenue; measures for WoW, MoM and YoY growth

Interactive dashboards
Two report pages with week, quarter and gender slicers, plus insights and recommendations

Loading data into PostgreSQL

STEP 1 | CREATE DATABASE AND TABLES

```
-- create database and tables
CREATE DATABASE ccdb;

CREATE TABLE cc_detail (
  Client_Num INT,
  Card_Category VARCHAR(20),
  Week_Start_Date DATE,
  Credit_Limit DECIMAL(10,2),
  Total_Trans_Amt INT,
  Interest_Earned DECIMAL(10,3),
  -- 18 columns in total
);

CREATE TABLE cust_detail (
  Client_Num INT,
  Customer_Age INT,
  Income INT,
  -- 15 columns in total
);
```

STEP 2 | LOAD CSV FILES WITH COPY

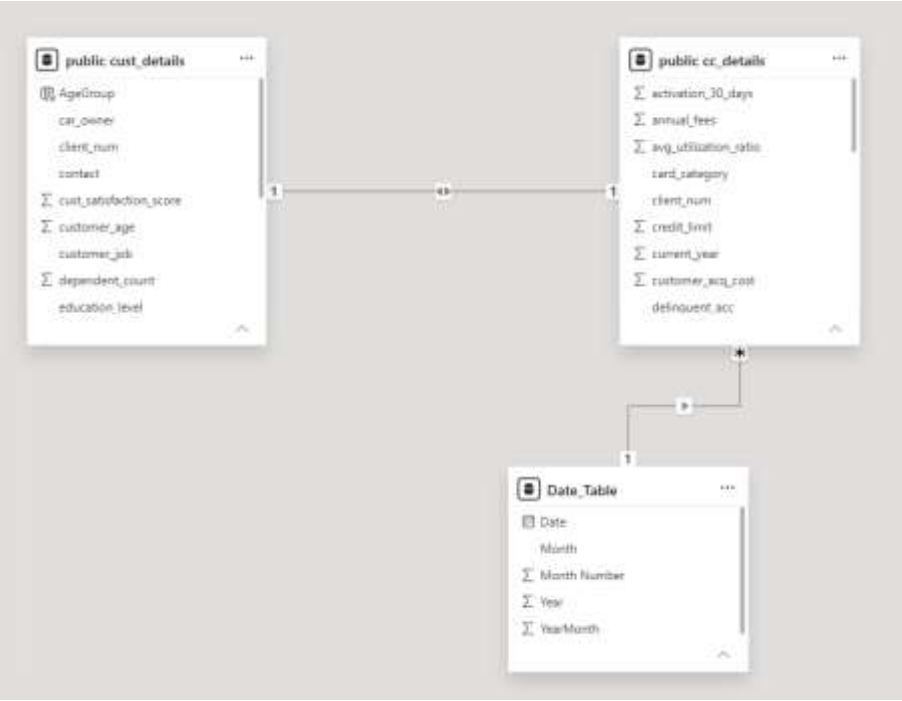
```
-- load the main data
COPY cc_detail
FROM 'D:\credit_card.csv'
DELIMITER ','
CSV HEADER;

-- fix date format error
SET datestyle TO 'ISO, DMY';

-- append week-53 data
COPY cc_detail
FROM 'D:\cc_add.csv'
DELIMITER ','
CSV HEADER;
```

Step 3: Power BI connects to ccdb and imports the tables

Data model



cc_details | Fact
Fees, interest, transactions. One-to-one with customers on client_num

cust_details | Dimension
Age, income, state, education, 10,108 customers

Date_Table | Calendar
358 days from Jan 2023, one-to-many to card data

DAX: calculated columns and tables

CALCULATED COLUMN | cust_details

```
AgeGroup = SWITCH(TRUE(),
  [customer_age] < 30, "20-30",
  [customer_age] >= 30 && [customer_age] < 40, "30-40",
  [customer_age] >= 40 && [customer_age] < 50, "40-50",
  [customer_age] >= 50 && [customer_age] < 60, "50-60",
  [customer_age] >= 60, "60+",
  "unknown")
```

CALCULATED COLUMNS | cc_details

```
Revenue =
  [annual_fees]
  + [total_trans_amt]
  + [interest_earned]

week_num2 =
  WEEKNUM([week_start_date])
```

CALCULATED COLUMN | cust_details

```
Income_Group = SWITCH(TRUE(),
  [income] < 35000, "Low",
  [income] >= 35000 && [income] < 70000, "Med",
  [income] >= 70000, "High",
  "Unknown")
```

CALCULATED TABLE | Date_Table

```
ADDCOLUMNS(
  CALENDAR(
    MIN([week_start_date]),
    MAX([week_start_date])),
  "Year", YEAR([Date]),
  "Month Number", MONTH([Date]),
  "Month", FORMAT([Date],"MMM"),
  "YearMonth",
    YEAR([Date])*100+MONTH([Date]))
```

DAX: measures and time intelligence

MEASURES | latest-week snapshot

```
TotalRevenue = SUM(cc_details[Revenue])

Current_Week_Revenue =
  CALCULATE(SUM(cc_details[Revenue]),
    FILTER(ALL(cc_details),
      cc_details[week_num2] =
        MAX(cc_details[week_num2])))

Previous_week_revenue =
  CALCULATE(SUM(cc_details[Revenue]),
    FILTER(ALL(cc_details),
      cc_details[week_num2] =
        MAX(cc_details[week_num2]) - 1))
```

MEASURES | week-over-week growth

```
Previous_week's_revenue =
  CALCULATE([TotalRevenue],
    DATEADD(Date_Table[Date], -7, DAY))

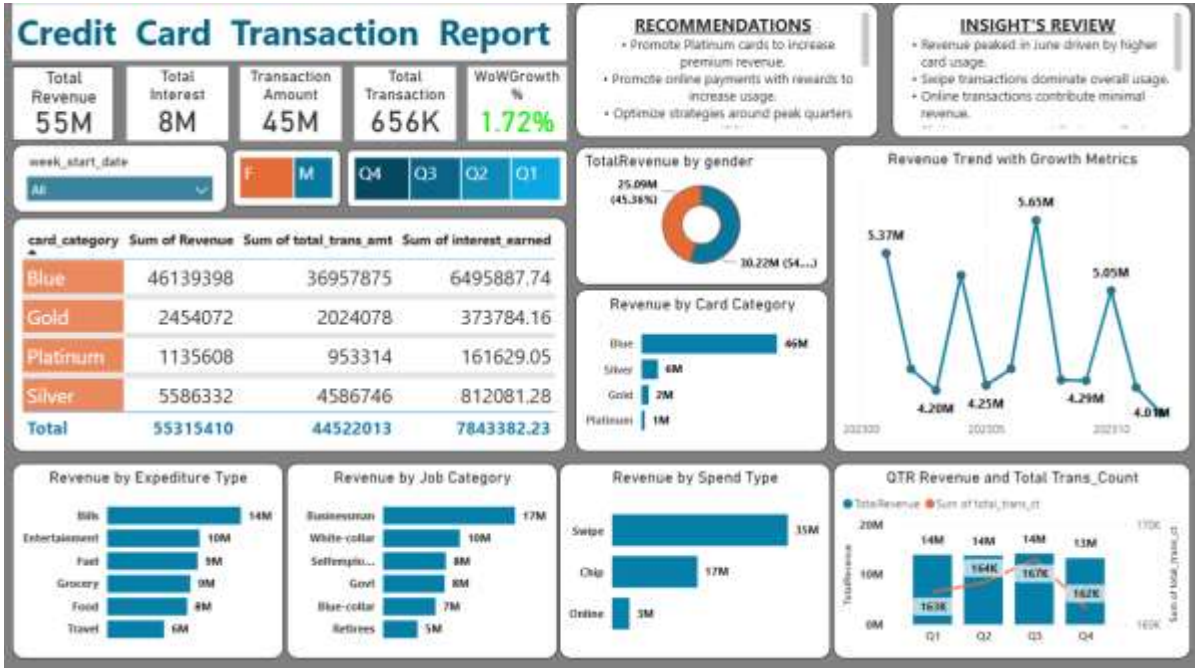
WoW_growth% =
  DIVIDE([TotalRevenue]
    - [Previous_week's_revenue],
    [Previous_week's_revenue], 0)
```

Same pattern, new offset

MoM_Growth%: DATEADD(..., -1, MONTH)

YoY_Growth%: DATEADD(..., -1, YEAR)

Dashboard 1: transaction report



KPIs, card and channel breakdowns, quarterly and growth trends

1.72%

Week-over-week revenue growth

~63%

of revenue from card swipes

83%

of revenue from Blue cards

Dashboard 2: customer report

~13M

revenue each from TX, NY, CA

~29M

from high-income (~52%)

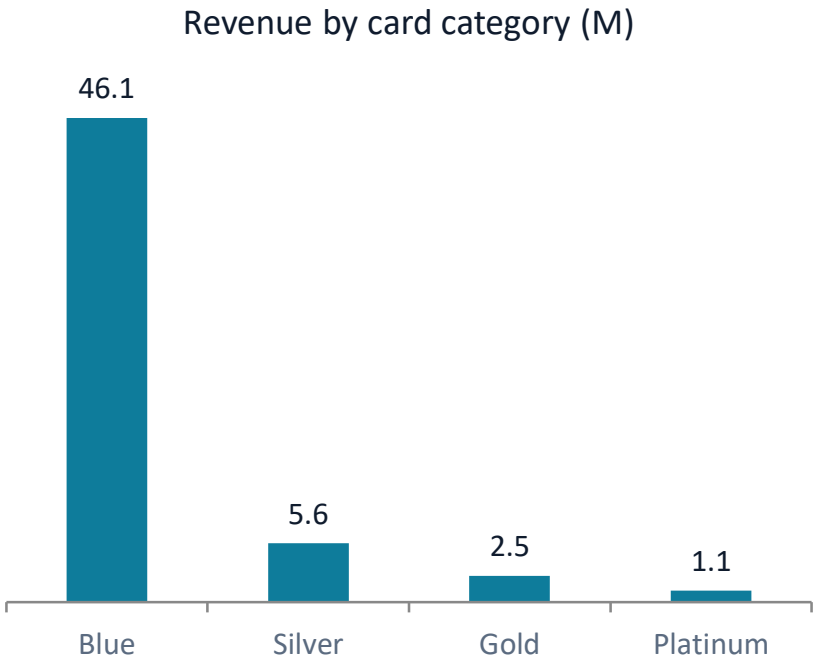
40-50

age band earns most (~23M)



Revenue by income, state, education, age and marital status

Key insights



1. Blue and Silver cards drive 93.5% of revenue; Platinum adds only 2%.
2. Swipe transactions lead at 35M; online adds just 3M.
3. Male customers bring 30.2M (54.6%) against 25.1M from female customers.
4. Quarterly revenue is steady near 14M, with Q4 easing to 13M.

Recommendations and takeaways

Recommendations

- Promote Platinum cards to lift premium revenue
- Reward online payments to grow an under-used channel
- Target high-spend, middle-aged (40-50) segments
- Plan campaigns around peak quarters

Skills demonstrated



SQL data sourcing



Data modelling and relationships



DAX: SWITCH, CALCULATE, DATEADD



Dashboard design and storytelling